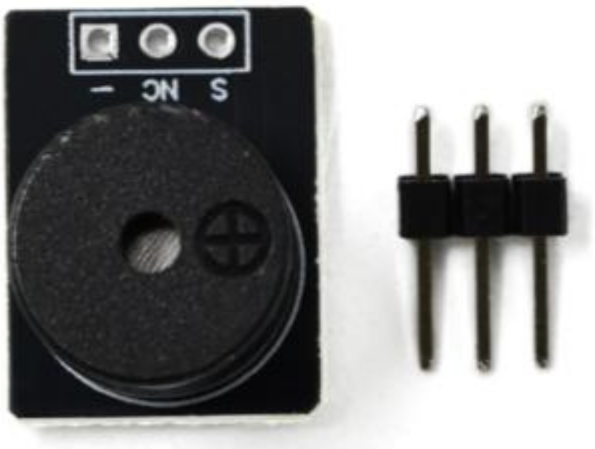
SmartElex Passive Buzzer Module



The **SmartElex Passive Buzzer Module** is a simple sound-making module you can use with microcontrollers like **Arduino**, **ESP32**, **Raspberry Pi**, and others for audible feedback in your projects.

📌 What It Is

A **passive buzzer** doesn't have a built-in oscillator — instead, it requires an **external signal (PWM or frequency output)** from your microcontroller to produce sound. This means you can generate **variable tones and even simple melodies**.

🔋 Key Specs

Specification	Value
Operating Voltage	~3.3 V to 5 V DC (Arduino/ESP32 friendly)
Tone Range	~1.5 kHz to 2.5 kHz depending on drive frequency
Interface	Digital pin (PWM)
Mounting	Standard 2.54 mm pin header, sometimes with bolt holes
Material / Build	Lightweight plastic module

🔌 Typical Pins & Connections

Buzzer Pin	Connects To	Function
VCC	5 V (Arduino) / 3.3 V (ESP32)	Power
GND	GND	Common ground
SIG / IN	Digital I/O	PWM or digital signal to generate tones

⚠️ Ensure a common ground between your microcontroller and the buzzer module.

🧠 How It Works

Unlike an active buzzer that produces a fixed tone when powered, a **passive buzzer** needs you to send a **square-wave signal (PWM)** at a specific frequency to create sound. That lets you control:

✓ Tone pitch
✓ Duration
✓ Simple melodies

For example, driving at **2 kHz** yields a tone around that audible frequency range.

📌 Example: Arduino PWM Tone

Here's a simple sketch to make the passive buzzer play a sound using Arduino's tone() function:

```
#define BUZZER_PIN 8

void setup() {
  pinMode(BUZZER_PIN, OUTPUT);
}

void loop() {
  tone(BUZZER_PIN, 2000); // 2 kHz tone
  delay(500);
  noTone(BUZZER_PIN);
  delay(500);
}
```

✓ Produces a beep every second.

✓ Works because tone() generates a PWM frequency that drives a passive buzzer.

📌 Example: ESP32 PWM (No tone())

On ESP32, use the LEDC PWM system:

```
#define BUZZER_PIN 23
const int channel = 0;
const int freq = 2000;
const int resolution = 8; // 8-bit

void setup() {
  ledcSetup(channel, freq, resolution);
  ledcAttachPin(BUZZER_PIN, channel);
}

void loop() {
  ledcWrite(channel, 128); // PWM on
  delay(500);
  ledcWrite(channel, 0);   // PWM off
  delay(500);
}
```

✓ This generates a tone at ~2 kHz.

✓ You can vary freq to change pitch.

📌 Typical Use Cases

✅ Alarm or alert sounds

✅ Beeps for button feedback

✅ Simple melodies or UI tones

✅ Timers and notifications in projects

✅ When to Use Passive vs Active

Feature	Passive Buzzer	Active Buzzer
Built-in tone generator	❌	✅
Variable tones	✅ (via PWM)	❌ (fixed)
Easy control	⚠ (needs signal)	✓ (just ON/OFF)
Simple beeps	✓	✓

Passive buzzers are ideal when you want **different tones or melodies**; active buzzers are best for simple beeps.